

SEMESTER S8

SECURE COMMUNICATION

Course Code	OEEVT 834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand network security services and mechanisms and the types of attacks

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	OSI security architecture, Security attacks – Passive attacks, Active attacks, Security services- Authentication, Access Control, Data Confidentiality, Data integrity, Nonrepudiation, Availability service. Model for network security. Symmetric cipher model, Cryptography, Crypto analysis, Substitution techniques- Hill Cipher, One time pad, Transposition Techniques	9
2	Finite Fields Groups, Rings and Fields, Modular arithmetic, Euclidian algorithm, Finite Fields of the form GF(p), Polynomial arithmetic	8
3	Block Cipher Principles – Stream Ciphers and Block Ciphers, Feistel Cipher, Feistel Decryption algorithm, The Data encryption standard, DES Decryption - Avalanche effect, The AES Cipher, substitute bytes transformation, Shift row transformation, Mix Column transformation	9
4	Public Key Cryptography, RSA and Key Management Principles of public key cryptosystems-Public key cryptosystems, Application for Public key cryptosystem requirements, Fermat's theorem, Euler's Totient Function, Euler's theorem, RSA algorithm, Key management, Distribution of public keys, Publicly available directory, Public key authority, public key certificates, Distribution of secret keys using public key cryptography, Public Key Encryption, Message Authentication Code, Hash function	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate network security services, mechanisms and the types of attacks and also to understand symmetric encryption process and different encryption techniques	K2
CO2	Apply the concepts of group, ring, field, modular arithmetic, Euclidean algorithm, Finite fields and polynomial arithmetic	K3
CO3	Outline the principles of modern symmetric ciphers like the Data Encryption Standard and Advanced Encryption Standard	K2
CO4	Describe the concepts of public key cryptography, RSA algorithm, key distribution and management for public key systems	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	3										2
CO3	3	2										2
CO4	3	2										2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cryptography and Network security: principles and practice	William Stallings	Prentice Hall of India	4 th Edition, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cryptography and Network security Tata McGraw-Hill,	Behrouz A. Forouzan	Tata McGraw-Hill	2008
2	Abstract Algebra	David S. Dummit & Richard M Foote	Wiley India Pvt. Ltd.	2 nd Edition, 2008
3	Cryptography, Theory and Practice	Douglas A. Stinson	Chapman & Hall, CRC Press Company, Washington	2005
4	Elliptic Curves: Theory and Cryptography	Lawrence C. Washington	Chapman & Hall, CRC Press Company, Washington	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105162/ CRYPTOGRAPHY AND NETWORK SECURITY, PROF. SOURAV MUKHOPADHYAY Department of Computer Science and Engineering IIT Kharagpur
2	https://archive.nptel.ac.in/courses/106/105/106105162/ CRYPTOGRAPHY AND NETWORK SECURITY, PROF. SOURAV MUKHOPADHYAY Department of Computer Science and Engineering IIT Kharagpur
3	https://archive.nptel.ac.in/courses/106/105/106105162/ CRYPTOGRAPHY AND NETWORK SECURITY, PROF. SOURAV MUKHOPADHYAY Department of Computer Science and Engineering IIT Kharagpur
4	https://archive.nptel.ac.in/courses/106/105/106105162/ CRYPTOGRAPHY AND NETWORK SECURITY, PROF. SOURAV MUKHOPADHYAY Department of Computer Science and Engineering IIT Kharagpur